

Alexander Steinmaurer - Publication List

Updated 09 September 2026

Peer-reviewed Publications

2026

- Cipriano, B. P., Petrovska, O., Pombo, N., Battestilli, L., Farinetti, L., Glassey, R., Kasinidou, M., Olayinka, O., Shah, A., Steinmaurer, A., Vaidhiyanathan, R., & Weichert, J. (2026). Towards Improving CS Students' Generative AI Literacy. In *Proceedings of the 31st ACM Conference on Innovation and Technology in Computer Science Education V. 2* (pp. 789-790). ACM. <https://doi.org/10.1145/3803401.3812055>
- Gürtl, S., Maier, J., Scharf, D., Gugacs, M., Gütl, C., & Steinmaurer, A. (2026). Supporting Tutors in Assessing Student Software Project Ideas with LLM-Based Pre-assessments. In *Innovation via Collaborative Learning in Engineering Education* (pp. 207-218). Springer. https://doi.org/10.1007/978-3-032-18888-5_20
- Köroglu, M. N., Mayrhofer, J., Steinmaurer, A., Wintersberger, P., Dennerlein, S. M., & Weinhandl, R. (2026). AI-Supported Lecturer Decision-Making in Higher Education: A Socio-Technical Perspective. *International Journal of Educational Technology in Higher Education*, 23(1), 49. <https://doi.org/10.1186/s41239-026-00623-8>

2025

- Bouvier, D. J., Cipriano, B. P., Glassey, R., Petrovska, O., Anderson, E., Birillo, A., Dougherty, R. E., Pettit, R., Pombo, N., Rahimi, E., Ramakrishnan, C., Steinmaurer, A., Taneja, S., Usman, M., & Vadaparty, A. (2025). The Rest of the Robots: Generative AI in Post-introductory Computing Education. In *Proceedings of the 2025 Working Group Reports on Innovation and Technology in Computer Science Education* (pp. 61-107). ACM. <https://doi.org/10.1145/3760545.3783970>
- Bouvier, D. J., Cipriano, B. P., Glassey, R., Pettit, R., Anderson, E., Birillo, A., Dougherty, R. E., Hazzan, O., Petrovska, O., Pombo, N., Rahimi, E., Ramakrishnan, C., Steinmaurer, A., Taneja, S., Usman, M., Vadaparty, A., & Yeluripati, G. R. (2025). GenAI Integration in Upper-Level Computing Courses. In *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 2* (pp. 691-692). ACM. <https://doi.org/10.1145/3724389.3731276>
- Gast, S., Felix, S. D., Steinmaurer, A., Juffinger, J., & Gruss, D. (2025). Real-World Study of the Security of Educational Test Systems. In *IEEE European Symposium on Security and Privacy Workshops (EuroS&P Workshops)* (pp. 588-599). IEEE. <https://doi.org/10.1109/EUROSPW67616.2025.00074>
- Gürtl, S., Scharf, D., Thrainer, C., Gütl, C., & Steinmaurer, A. (2025). Design and Evaluation of an LLM-Based Mentor for Software Architecture in Higher Education Project Management Classes. In *Futureproofing Engineering Education for Global Responsibility* (pp. 375-386). Springer. https://doi.org/10.1007/978-3-031-83523-0_35
- Kerschbaumer, D., Schatz, C., Rupprechter, T., Gütl, C., & Steinmaurer, A. (2025). Do Comments Matter? Investigating Students' Source Code Comment Behaviour and Its Relation to Academic Success in a CS1 Course. In *Futureproofing Engineering Education for Global Responsibility* (pp. 519-530). Springer. https://doi.org/10.1007/978-3-031-85649-5_51
- Schnedlitz, F., Kerschbaumer, D., Steinmaurer, A., Lowe, D., & Gütl, C. (2025). To Complete or Not to Complete: Prediction and Classification of Dropouts in a CS1 Course. In *IEEE Global Engineering Education Conference (EDUCON)* (pp. 1-7). IEEE. <https://doi.org/10.1109/EDUCON62633.2025.11016432>

2024

- Bajramovic, A., Steinmaurer, A., & Gütl, C. (2024). Playing Video Games to Learn Computational Thinking? A Case Study of Game-Based Learning in Secondary CS Education. In *Towards a Hybrid, Flexible and Socially Engaged Higher Education* (pp. 474-482). Springer. https://doi.org/10.1007/978-3-031-53022-7_47

- Kerschbaumer, D., Steinmaurer, A., & Gütl, C. (2024). A Code-Driven Exploration of Key C Language Concepts in a CS1 Class. In *Smart Mobile Communication & Artificial Intelligence* (pp. 397-408). Springer. https://doi.org/10.1007/978-3-031-54327-2_40
- Malkoc, S., Steinmaurer, A., Gütl, C., Luttenberger, S., & Paechter, M. (2024). Coding Decoded: Factors Related to Achievement and Gender Differences in an Online Flipped Classroom Programming University Course. *Education Sciences*, 14(6), Article 634. <https://doi.org/10.3390/educsci14060634>
- Pester, A., Tammaa, A., Gütl, C., Steinmaurer, A., & Abou El-Seoud, S. (2024). Conversational Agents, Virtual Worlds, and Beyond: A Review of Large Language Models Enabling Immersive Learning. In *IEEE Global Engineering Education Conference (EDUCON)* (pp. 1-6). IEEE. <https://doi.org/10.1109/EDUCON60312.2024.10578895>
- Rivera, E., Steinmaurer, A., Fisler, K., & Krishnamurthi, S. (2024). Iterative Student Program Planning using Transformer-Driven Feedback. In *Proceedings of the 2024 ACM Conference on Innovation and Technology in Computer Science Education V. 1*. ACM. <https://doi.org/10.1145/3649217.3653607>
- Steinmaurer, A., Dengel, A., Comanici, M., Buchner, J., Memminger, J., & Gütl, C. (2024). Immersive Learning in History Education: Exploring the Capabilities of Virtual Avatars and Large Language Models. In *Immersive Learning Research Network - 10th International Conference on Immersive Learning* (pp. 363-374). Springer. https://doi.org/10.1007/978-3-031-80475-5_26 **[Best Short Paper Award]**

2023

- Dengel, A., Steinmaurer, A., Müller, L. M., Platz, M., Wang, M., Gütl, C., Pester, A., & Morgado, L. (2023). Research Agenda 2030: The Great Questions of Immersive Learning Research. In *Immersive Learning Research Network - 9th International Conference* (pp. 161-172). Springer. https://doi.org/10.1007/978-3-031-47328-9_12
- Rebol, M., Steinmaurer, A., Gamillscheg, F., Pietroszek, K., Gütl, C., Ranniger, C., Hood, C., Rutenberg, A., & Sikka, N. (2023). CPR Emergency Assistance Through Mixed Reality Communication. In *Augmented Intelligence and Intelligent Tutoring Systems* (pp. 415-429). Springer. https://doi.org/10.1007/978-3-031-32883-1_38
- Steinmaurer, A., Savic, R., Krugel, J., & Gütl, C. (2023). Analyzing Social Networks and Learning Content on a Discussion Forum of an Introductory Programming MOOC in Higher Education. In *IEEE Learning with MOOCS* (pp. 1-6). IEEE. <https://doi.org/10.1109/LWMOOCS58322.2023.10305937> **[Best Paper Award]**

2022

- Rebol, M., Pietroszek, K., Ranniger, C., Hood, C., Rutenberg, A., Sikka, N., Steinmaurer, A., & Gütl, C. (2022). Work-in-Progress-Volumetric Communication for Remote Assistance Giving Cardiopulmonary Resuscitation. In *8th International Conference of the Immersive Learning Research Network* (pp. 1-3). IEEE. <https://doi.org/10.23919/ILRN55037.2022.9815989>
- Steinmaurer, A., & Gütl, C. (2022). Implementation and Experiences of a Flipped Lecture Hall - A Fully Online Introductory Programming Course. In *Learning in the Age of Digital and Green Transition* (pp. 832-843). Springer. https://doi.org/10.1007/978-3-031-26876-2_79
- Steinmaurer, A., Eckhard, D., Dreveny, J., & Gütl, C. (2022). Developing and Evaluating a Multiplayer Game Mode in a Programming Learning Environment. In *8th International Conference of the Immersive Learning Research Network* (pp. 1-8). IEEE. <https://doi.org/10.23919/ILRN55037.2022.9815973>
- Steinmaurer, A., Schatz, C., Krugel, J., & Gütl, C. (2022). Analyzing Behavioral Patterns in an Introductory Programming MOOC at University Level. In *IEEE Learning with MOOCS* (pp. 114-119). IEEE. <https://doi.org/10.1109/LWMOOCS53067.2022.9927886>
- Steinmaurer, A., Bajramovic, A., Pollhammer, D., & Gütl, C. (2022). Learning Security Awareness in Email Communication Using a Platform for Digital Skill Teaching. In *IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)* (pp. 38-45). IEEE. <https://doi.org/10.1109/TALE54877.2022.00015>

Steinmaurer, A., Mesarec, B., Mesarec, T., Pietroszek, K., & Gütl, C. (2022). Mobile XR Interface for Quantum Computing. In *Adjunct Proceedings of the 2022 ACM International Conference on Interactive Media Experiences* (pp. 243-248). <https://doi.org/10.6084/m9.figshare.20069621.v1>

2021

Pirker, J., Steinmaurer, A., & Karakas, A. (2021). Beyond Gaming: The Potential of Twitch for Online Learning and Teaching. In *Proceedings of the 26th ACM Conference on Innovation and Technology in Computer Science Education V. 1* (pp. 74-80). ACM. <https://doi.org/10.1145/3430665.3456324>

Steinmaurer, A., Sackl, M., & Gütl, C. (2021). Engagement in In-Game Questionnaires - Perspectives from Users and Experts. In *7th International Conference of the Immersive Learning Research Network* (pp. 1-7). IEEE. <https://doi.org/10.23919/ILRN52045.2021.9459373>

Steinmaurer, A., Tilanthe, A. K., & Gütl, C. (2021). Designing and Developing a Learning Analytics Platform for the Coding Learning Game sCool. In *New Realities, Mobile Systems and Applications* (pp. 547-558). Springer. https://doi.org/10.1007/978-3-030-96296-8_49

2020

Mosquera, C. K., Steinmaurer, A., Eckhardt, C., & Guetl, C. (2020). Immersively Learning Object Oriented Programming Concepts With sCool. In *6th International Conference of the Immersive Learning Research Network* (pp. 124-131). IEEE. <https://doi.org/10.23919/ILRN47897.2020.9155144>

Sackl, M., Steinmaurer, A., Cheong, C., Cheong, F., Filippou, J., & Gütl, C. (2020). sCool: Impact on Human-Computer Interface Improvements on Learner Experience in a Game-Based Learning Platform. In *Educating Engineers for Future Industrial Revolutions* (pp. 439-451). Springer. https://doi.org/10.1007/978-3-030-68198-2_41

2019

Steinmaurer, A., Pirker, J., & Gütl, C. (2019). sCool - Game-Based Learning in Computer Science Class: A Case Study in Secondary Education. *International Journal of Engineering Pedagogy*, 9(2), 35-50. <https://doi.org/10.3991/ijep.v9i2.9942>

2018

Steinmaurer, A., Pirker, J., & Gütl, C. (2018). sCool - Game Based Learning in STEM Education: A Case Study in Secondary Education. In *The Challenges of the Digital Transformation in Education* (pp. 614-625). Springer. https://doi.org/10.1007/978-3-030-11932-4_58

Book Chapters & Other Publications

2026

Steinmaurer, A., Kerschbaumer, D., & Gürtl, S. (2026). KI als Kollaborationspartner in der Informatiklehre. *fnma Magazin*, 43-45. <https://fnma.at/content/download/3484/23252?version=2>

2024

Müller, L. M., Domonkos, A., Steinmaurer, A., & Platz, M. (2024). MessbAR -- Entwicklung und Evaluation einer Augmented Reality App im Größenbereich Längen. In C. Bierbrauer, S. Ladel, & M. Platz (Eds.), *Lehren und Lernen mit digitalen Medien in der Primarstufe (PriMaMedien)*. WTM. <https://doi.org/10.37626/GA9783959873246.0>

2022

Steinmaurer, A., & Gütl, C. (2022). Computerspiele im Informatikunterricht. In H. Arndt (Ed.), *Digitale Spiele und fachliches Lernen* (pp. 211-247). FAU University Press. <https://doi.org/10.25593/978-3-96147-528-5>

Preprints & Manuscripts

2025

- Gürtl, S., Schimetta, G., Kerschbaumer, D., Liut, M., & Steinmaurer, A. (2025). *Automated Feedback on Student-Generated UML and ER Diagrams Using Large Language Models*. [Preprint]. CoRR.
<https://doi.org/10.48550/arXiv.2507.23470>
- Kerschbaumer, D., Ruprechter, T., Helic, D., Gütl, C., & Steinmaurer, A. (2025). *Individual Efforts in Group Programming Assignment*. [Unpublished manuscript].
- Liu, P., Lee, K. J. M., Steinmaurer, A., Picard-Deland, C., Carr, M., & Kitson, A. (2025). *DreamLLM-3D: Affective Dream Reliving using Large Language Model and 3D Generative AI*. [Preprint]. CoRR.
<https://doi.org/10.48550/arXiv.2503.16439>
- Pöschko, J., Steinmaurer, A., Gütl, C., Dennerlein, S., Paechter, M., & Kemény, F. (2025). *The Influence of Explainability and Source Credibility on Students' Trust in Large Language Model-Based Programming Learning Tools*. [Unpublished manuscript].

2023

- Steinmaurer, A., Dengel, A., Umfahrer, M., Zöhrer, K., Kogler, P., & Gütl, C. (2023). *A SMAwT Approach for Raising Social Media Awareness in Secondary CS Education -- Maybe a Little Bit Too Much?* [Unpublished manuscript].